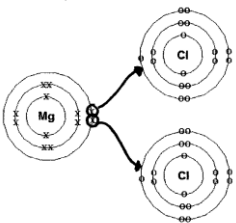


| Question |     |      |  | Marking details                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |  |  <p>both electrons must be transferred, both must go to the outer shells</p>                                               |                 | 1   |     | 1     |       |      |
|          |     | (ii) |  | <p>2,8 / (2,8)<sup>2+</sup> (1)</p> <p>– / 1– (1)                      neutral answer –1</p>                                                                                                                |                 | 2   |     | 2     |       |      |
|          | (b) | (i)  |  |  <p>must have shared pair and total of 8 electrons around both atoms</p> <p>ignore electrons in any inner shells drawn</p> |                 | 1   |     | 1     |       |      |
|          |     | (ii) |  | <p>covalent (bonding)</p> <p>reference to simple molecular / sharing electrons is neutral</p>                                                                                                               | 1               |     |     |       |       |      |
|          |     |      |  | Question 3 total                                                                                                                                                                                            | 1               | 4   | 0   | 5     | 0     | 0    |